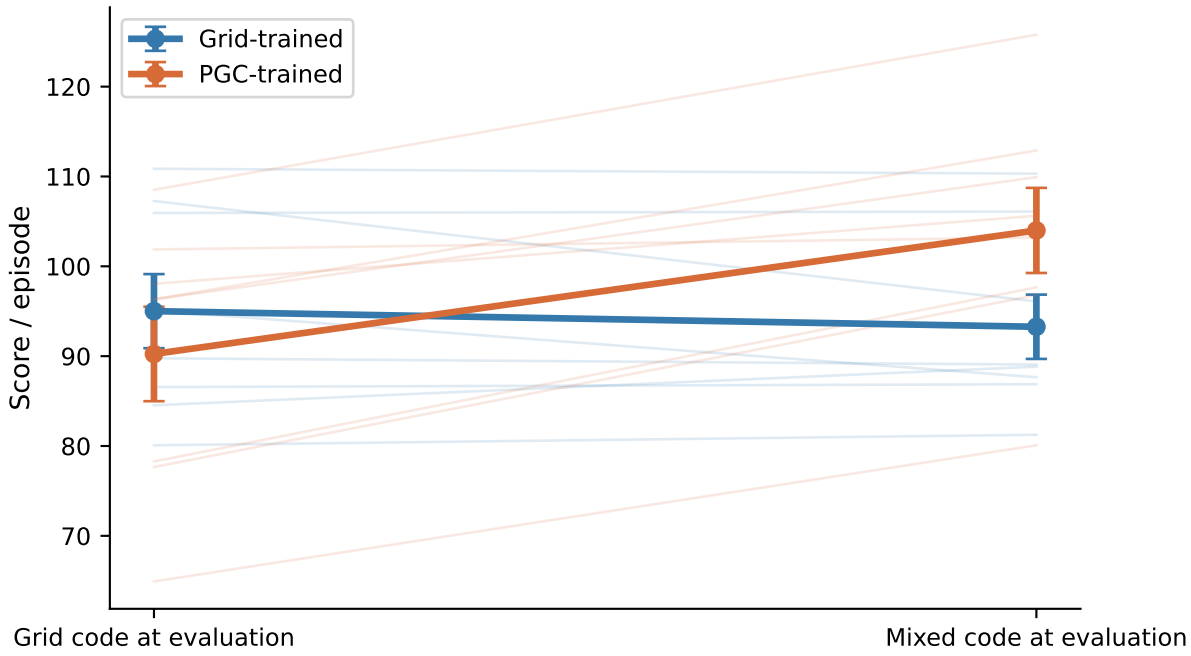
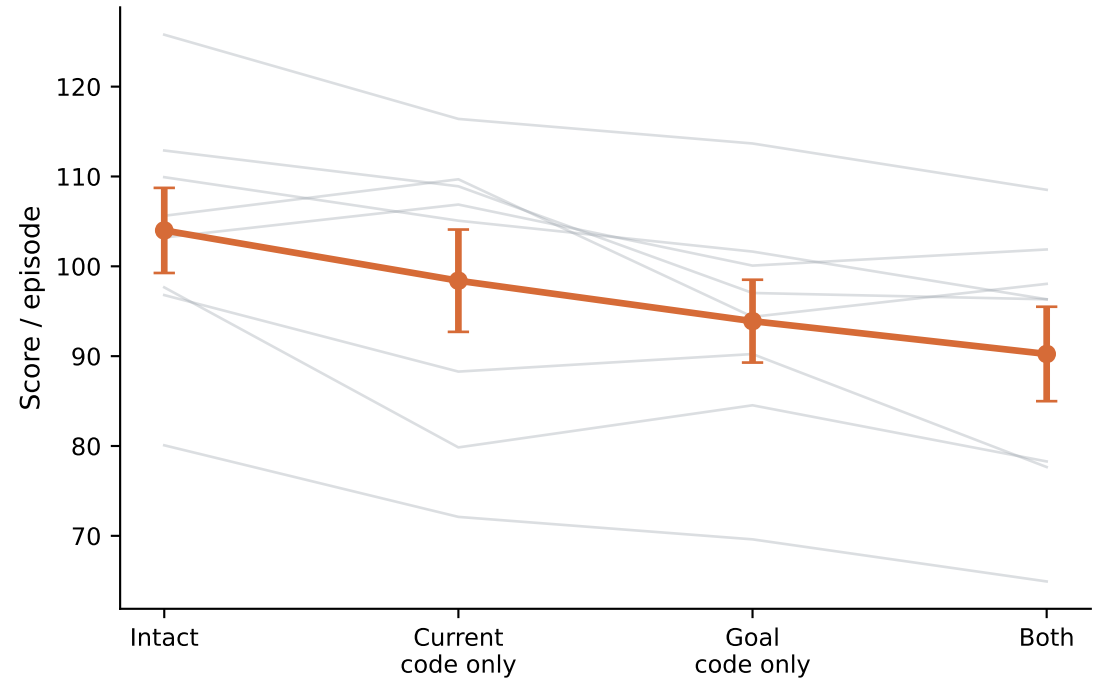
Frozen 1B LSTMs: population-code swaps and causal state interventions

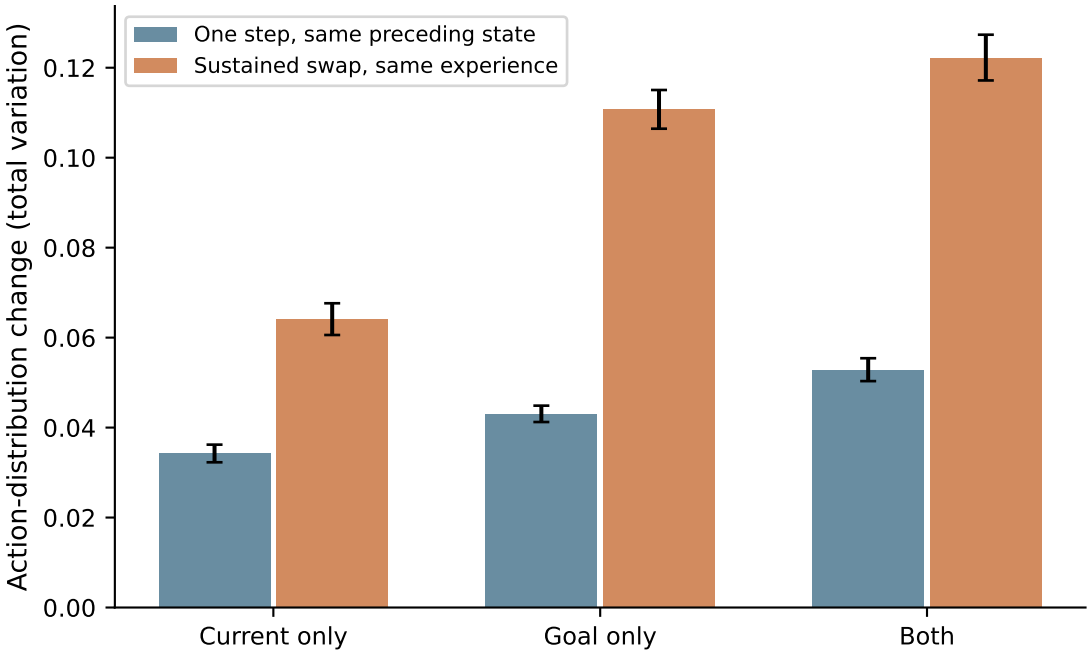
A | Same weights, different population code



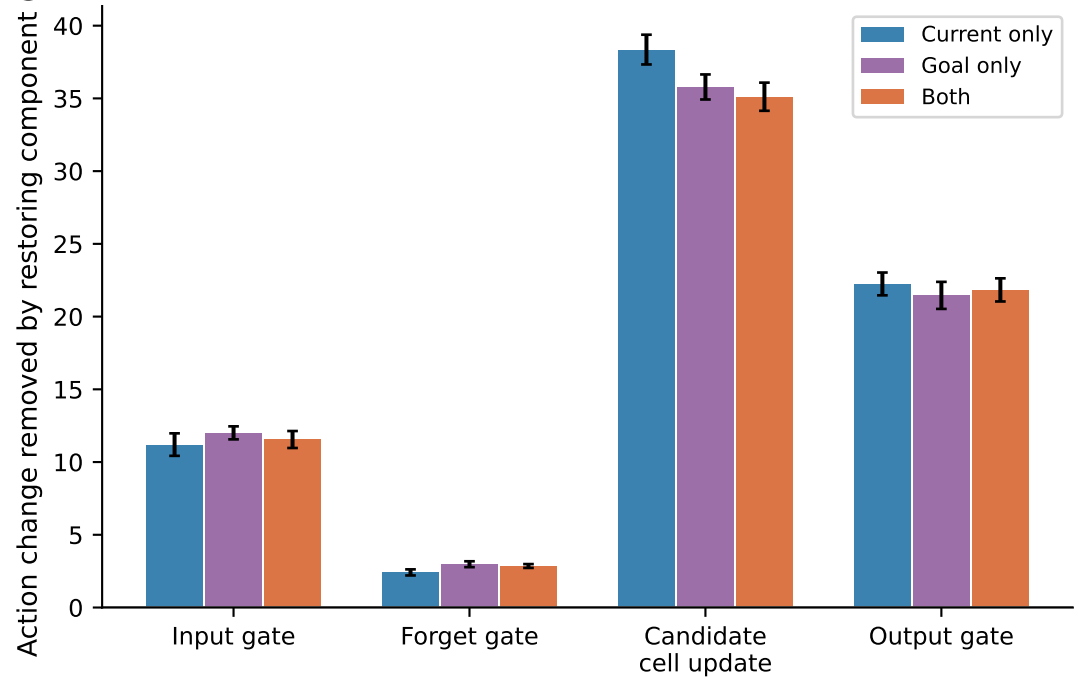
B | Which PGC input is replaced by grid code?



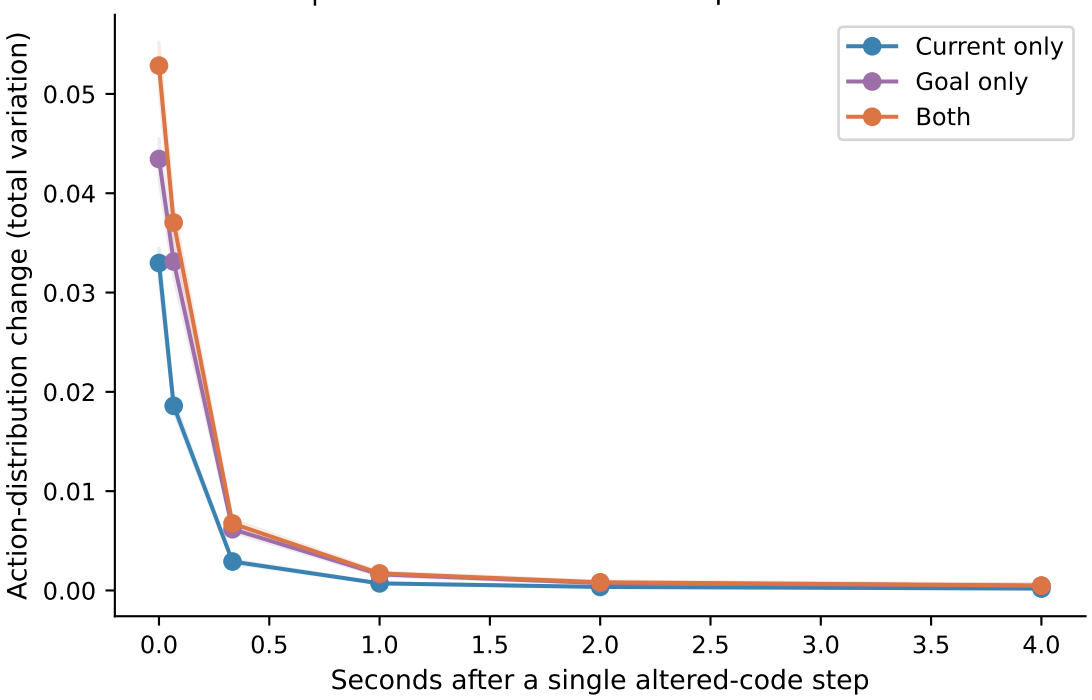
C | Internal effect before behavioral feedback



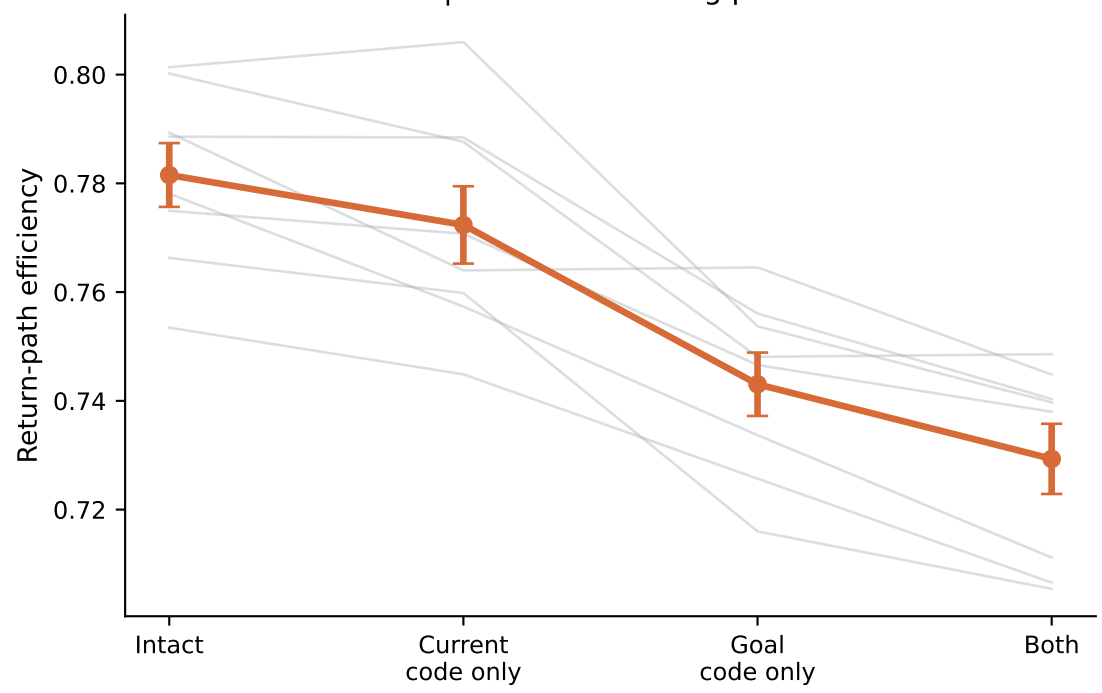
D | Causal restoration of one LSTM component



E | Persistence after intact input is restored



F | Effect on homing paths



Eight matched training seeds; 128 fresh episodes per condition per seed. Error bars / bands: SEM across seeds.
RNN forks share intact experience; gate-restoration effects interact and need not add to 100%.